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Navinshankar G V

Software Engineer / Backend Engineer

Portfolio: Navinshankar-Portfolio
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Experience

Software Engineer **March 2026 – Present**
TipTop Technologies *San Jose, California*

- Designed and maintained low-latency Java backend APIs for a real-time communication platform, cutting call-setup latency 25% and sustaining high-availability video and messaging workflows under production load
- Designed the session-handling layer for LiveKit video rooms to support concurrent room joins without blocking, cutting P99 room-join latency from 480ms to 210ms
- Built a fault-tolerant reconnect and room-recovery flow applying circuit-breaker and retry design patterns, cutting dropped-session rate from 3% to 0.5% and keeping calls stable through network drops and server restarts

Software Engineer Intern **February 2025 – May 2025**
IpserLab *San Jose, California*

- Built core backend services for a rental listing platform in Python and FastAPI, exposing async REST endpoints for search, listings, and bookings with Pydantic request and response validation
- Offloaded image processing, geocoding, and notifications to Celery workers, dropping listing-creation API response by 94% by returning as soon as jobs were queued
- Orchestrated containers on AWS ECS/Fargate with zero-downtime rolling deploys, reducing deploy-related incidents by 50%
- Automated Docker build & push to AWS ECR via GitHub Actions with layer caching, decreased build & release time by 85%

Software Developer **January 2022 – July 2023**
Viatore Solutions *Coimbatore, India*

- Decomposed the monolith into Java and Spring Boot microservices deployed independently on Kubernetes, isolating orders, inventory, and billing and cutting deployment lead time 75%.
- Optimized order-detail data access with Hibernate batch fetching & JPA projections cutting per-request database calls by 87%
- Engineered CI/CD pipelines using GitHub Actions to automate testing and deployment workflows, cutting release time by 75% and raising pre-production quality.
- Built order-management merchant dashboard with React and JavaScript, surfacing live order, stock, and shipment data for daily merchant operations into one view and replacing a manual, spreadsheet-based tracking process.
- Built a Kafka pipeline where an order-placed event fanned out to inventory, notifications, and analytics, processing 300 orders/day and decoupling checkout from downstream processing
- Replaced polling-based dashboard refresh with Redis Pub/Sub over WebSockets, reducing update latency by over 99% and cutting server CPU utilization by 40% by eliminating database bottlenecks
- Reduced initial dashboard bundle size by 45% and cut Time-to-Interactive by over 56% by implementing code-splitting and lazy-loading for heavy views.

Technical Skills

Languages: Java, JavaScript, SQL (PostgreSQL, MySQL), NoSQL (MongoDB, DynamoDB, Redis), Python, Bash, Linux
Frameworks & APIs: Spring Boot, Hibernate/JPA, Express.js, React, Angular, LiveKit,FastAPI, Django, REST, SOAP
Cloud & DevOps: Docker, Kubernetes, AWS (Lambda, SQS/SNS, Fargate, ECS/ECR), Terraform, GitHub Actions, GitLab CI/CD

Education

Master of Science in Information Systems, *Northeastern University, San Jose, California* **Dec. 2025**

Bachelor's of Engineering in Computer Science & Engineering, *Anna University, Chennai, India* **May 2022**

Projects

Research Summarizer | *Python, FastAPI, OpenAI API, Render, Docker* **December 2025**

- Built a RAG service that retrieves research papers from the arXiv API, ranks 20+ results per query to the top 5, and generates full-paper summaries; containerized with Docker and deployed on Render
- Designed automated test pipelines to validate LLM output on 50+ page inputs, applying confidence and relevance filters to flag low-confidence and off-topic summaries before they reached users
- Cut hallucination through prompt refinement and grounding checks, improving summary reliability on long-document inputs

HashtagHolidays | *Python, FastAPI, Docker, MongoDB, DialogFlow* **August 2025**

- Built a FastAPI-based recommendation service using Docker to process social interaction data, applying frequency–recency scoring, persisting ranked results in MongoDB, and exposing them through REST APIs and a conversational interface